

2-Mar-2025

- (1) BBox loss function
- (2) DETR
- (3) RT-DETR
- (4) Inference on RT-DETR
- (5) project Discussion

cspNet → DeNet

(1) BBox Loss function

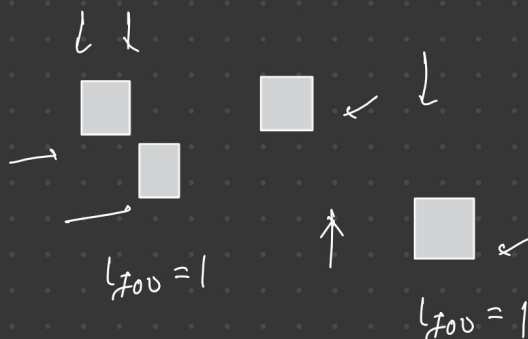
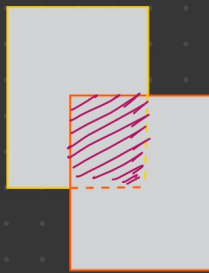
→ IOU

Advantage

→ Simple

Disadvantage

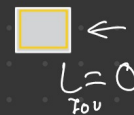
If boxes don't
touch the loss is
always 1



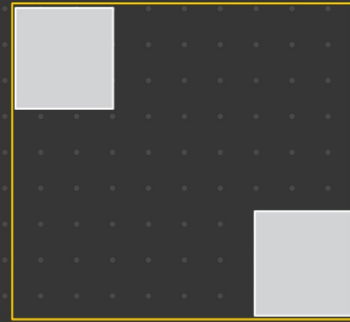
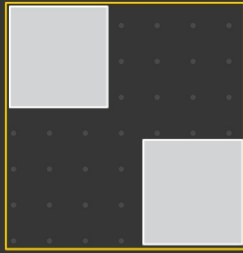
$$IOU = \frac{\text{Intersection Area}}{\text{Union Area}}$$

$$= \frac{4}{28} = \approx \underline{\underline{0.14}}$$

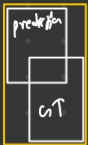
$$\begin{aligned} L_{loss} &= 1 - IOU \\ &= 1 - 0.14 \\ &= \approx \underline{\underline{0.86}} \end{aligned}$$



c₁₀₀ → Generalize IOU



$$c_{100} = \text{IOU} - \frac{\text{Enclosing Area} - \text{Union Area}}{\text{Enclosing area}}$$



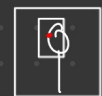
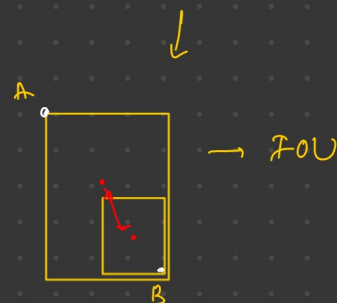
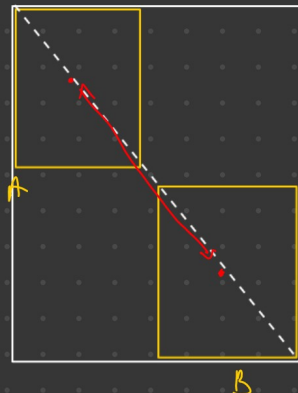
$$= 0.14 - \frac{36 - 28}{36} = 0.14 - 0.22 = \underline{\underline{-0.08}}$$

$$c_{100} = 1 - (-0.08) = \underline{\underline{1.08}}$$

— Advantage

— Encourages box to move closer, even if they don't touch.

(3) DIOS (Distance IOU)



1) Distance between centers:

$$d(B_p, B_t) \rightarrow \sqrt{(6-4)^2 + (6-4)^2} = \sqrt{8} \approx \underline{\underline{2.83}}$$

$B_p \rightarrow$ predicted BBox

$B_t \rightarrow$ target BBox

$$\begin{array}{l} B_p \rightarrow 4,4 \\ B_t \rightarrow 6,6 \end{array} \rightarrow \begin{array}{cc} \rightarrow [2,2] [6,6] \\ [4,4] [8,8] \leftarrow \end{array}$$

(2) Diagonal

$$\rightarrow \sqrt{(8-2)^2 + (8-2)^2} = \sqrt{72} = \underline{\underline{8.49}}$$

$$DIOU = IOU - \frac{d(B_p, B_t)^2}{d_{\max}^2} = \frac{(\sqrt{2})^2}{(\sqrt{72})^2}$$

$$= 0.14 - \frac{2 \cdot 2.83}{8.49} = 0.14 - 0.11 = 0.03$$

$$L_{DIOU} = \underline{\underline{1 - 0.03 = 0.97}}$$

(4) CIoU (Complete IoU)

Aspect ratio

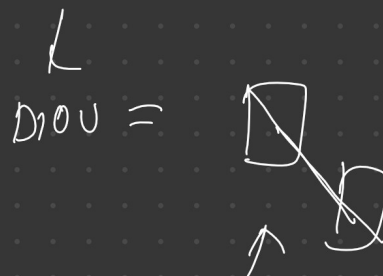
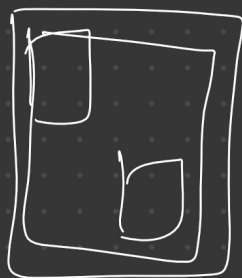
$$v = \frac{4}{\pi^2} \left(\arctan\left(\frac{w_t}{h_t}\right) - \arctan\left(\frac{w_p}{h_p}\right) \right)^2$$

$v=0$, for square boxes

$$CIoU = IOU - \frac{d(B_p, B_t)^2}{d_{max}^2} + \alpha v$$

$$\alpha = \frac{v}{(1 - IOU) + v}$$

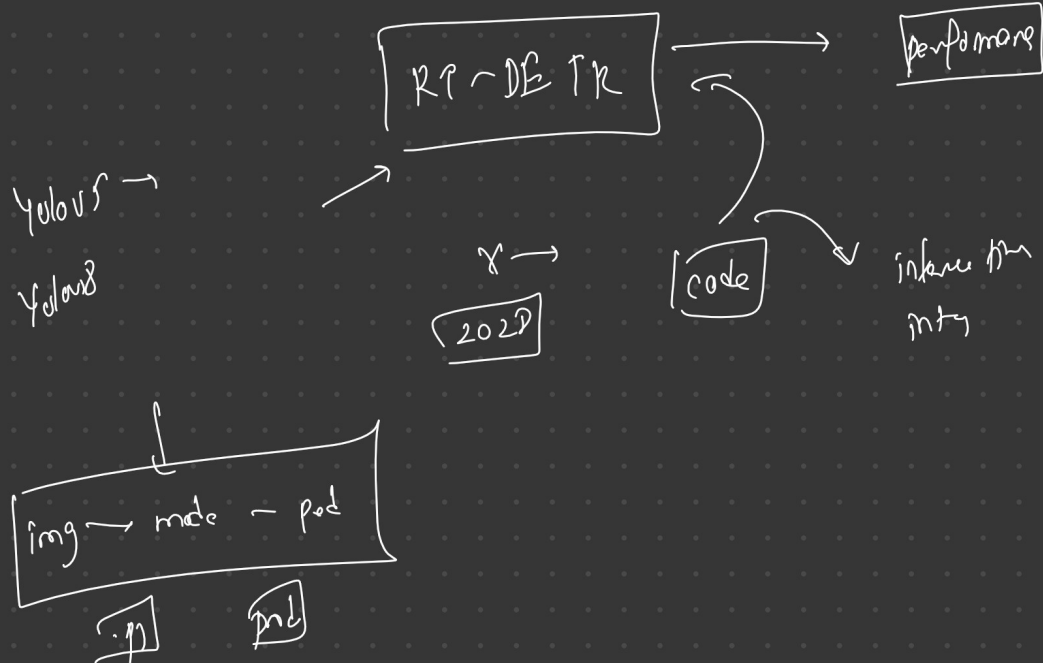
↓ ↓
IOU →
CIoU →



CIoU = aspect ratio

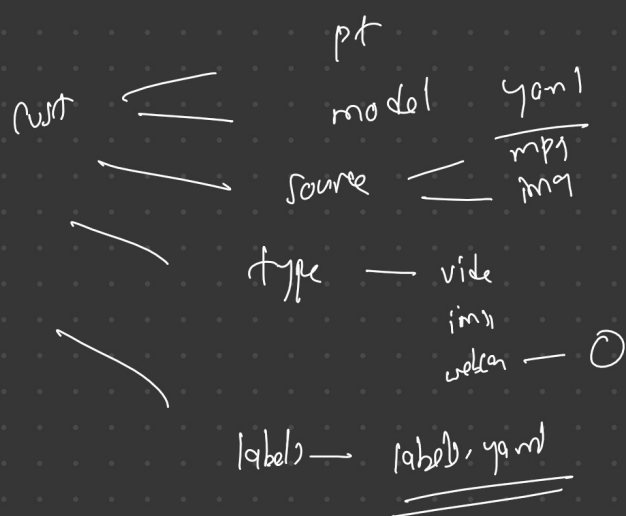


Overlap + Distance + Aspect ratio



RT-DETR

- Step 1 : Clone
- Step 2 : follow instructions.txt
- Step 3 : Download weights & config
- Step 4 : Copy the config path from the config folder
- Step 5 : Copy the model weight file from download location



Yolov5 →
 coco : yam
 train
 val
 test

label1 :
 ↓
 label1, yam1
point - label1

